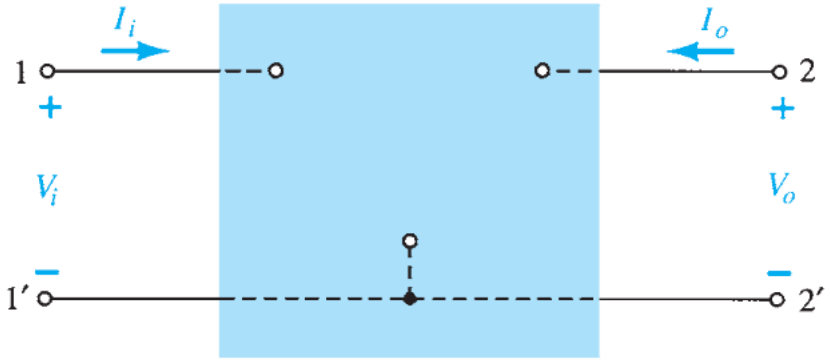
Hybrid Equivalent Model and Feedback Amplifiers

Ginanjar S. Adi

THE HYBRID EQUIVALENT MODEL (Review)

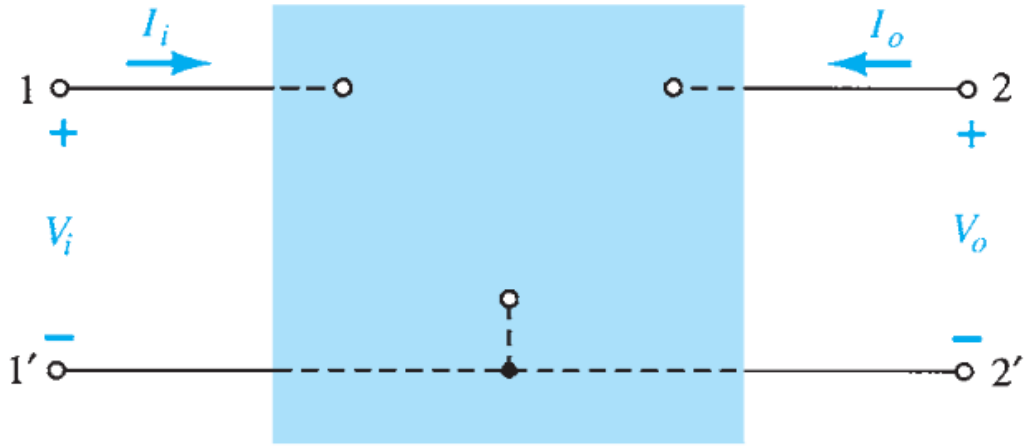
The parameters of the hybrid equivalent circuit are defined in general terms for any operating conditions.

- In other words, the hybrid parameters may not reflect the actual operating conditions but simply provide an indication of the level of each parameter to expect for general use.
- The description of the hybrid equivalent model will begin with the general two-port system of figure bellow:



THE HYBRID EQUIVALENT MODEL (Cont…)

The parameters relating the four variables are called *h-parameters*, from the word "hybrid." The term *hybrid* was chosen because the mixture of variables (*V* and *I*) in each equation results in a "hybrid" set of units of measurement for the *h*-parameters.



$$V_i = h_{11}I_i + h_{12}V_o$$

$$I_o = h_{21}I_i + h_{22}V_o$$

THE HYBRID EQUIVALENT MODEL (Cont…)

$$V_i = h_{11}I_i + h_{12}V_o$$

- **h_{11}** If we arbitrarily set V_o = 0 (short circuit the output terminals) and solve for h_{11}, we find (*short-circuit input-impedance parameter*)

$$h_{11} = \left.\frac{V_i}{I_i}\right|_{V_o=0} \quad \text{ohms}$$

- **h_{12}** If I_i is set equal to zero by opening the input leads, the following results for h_{12} (*open-circuit reverse transfer voltage ratio parameter*)

$$h_{12} = \left.\frac{V_i}{V_o}\right|_{I_i=0} \quad \text{unitless}$$

THE HYBRID EQUIVALENT MODEL (Cont…)

$$I_o = h_{21}I_i + h_{22}V_o$$

- ***h_{21}*** If V_o is set equal to zero by again shorting the output terminals, the following results for h_{21} (*short- circuit forward transfer current ratio parameter*)

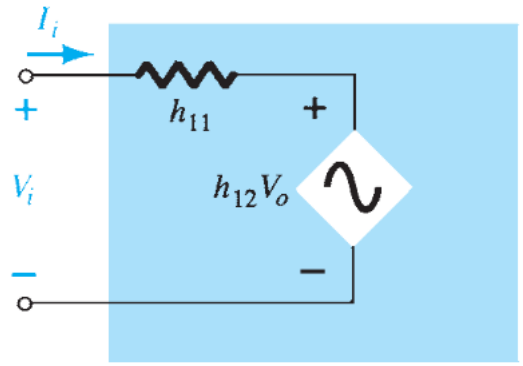
$$h_{21} = \left.\frac{I_o}{I_i}\right|_{V_o=0}$$ unitless

- ***h_{22}*** can be found by again opening the input leads to set $I_1 = 0$ and solving for h_{22} (*open-circuit output admittance parameter*)

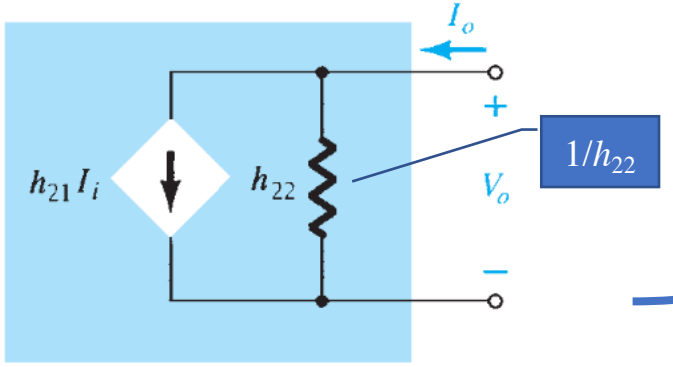
$$h_{22} = \left.\frac{I_o}{V_o}\right|_{I_i=0}$$ siemens

THE HYBRID EQUIVALENT MODEL (Cont…)

- KVL (Input)



- KCL (Output)

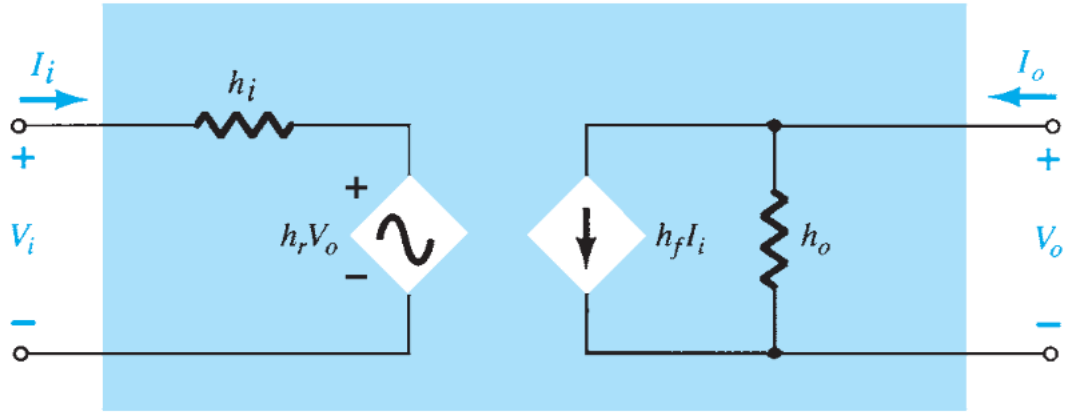


$h_{11} \rightarrow$ *i*nput resistance $\rightarrow h_i$

$h_{12} \rightarrow$ *r*everse transfer voltage ratio $\rightarrow h_r$

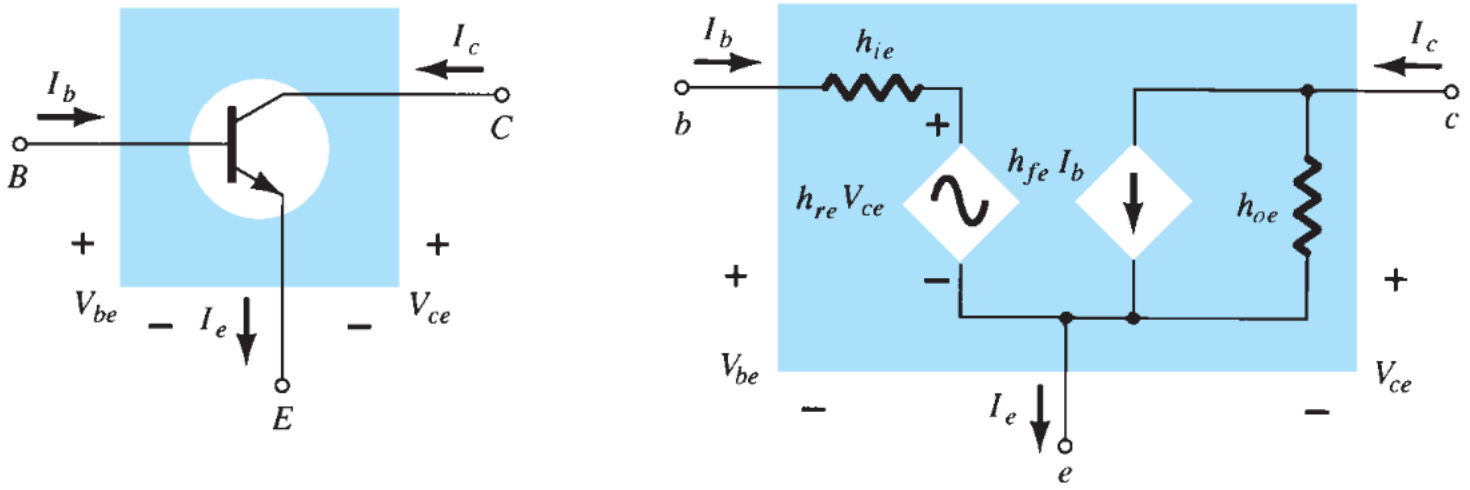
$h_{21} \rightarrow$ *f*orward transfer current ratio $\rightarrow h_f$

$h_{22} \rightarrow$ *o*utput conductance $\rightarrow h_o$



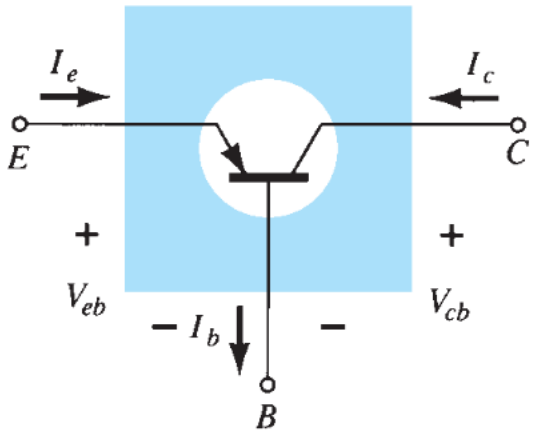
THE HYBRID EQUIVALENT MODEL (Cont…)

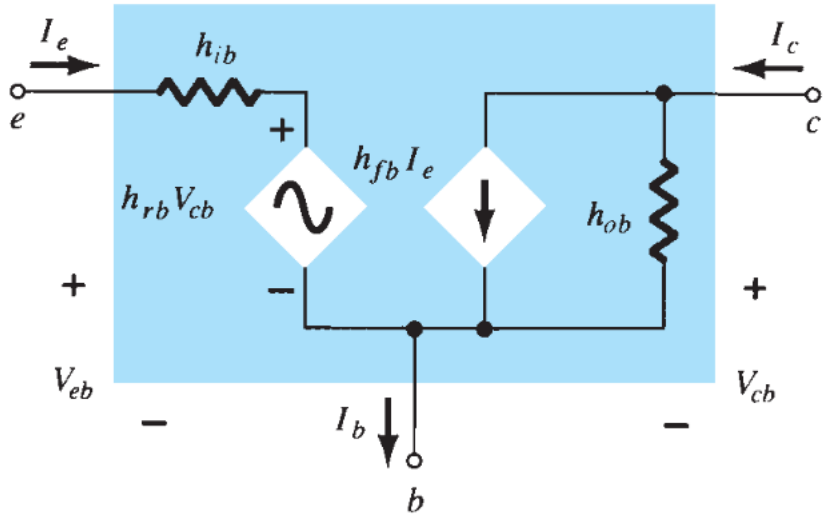
- Common-emitter configuration:



THE HYBRID EQUIVALENT MODEL (Cont…)

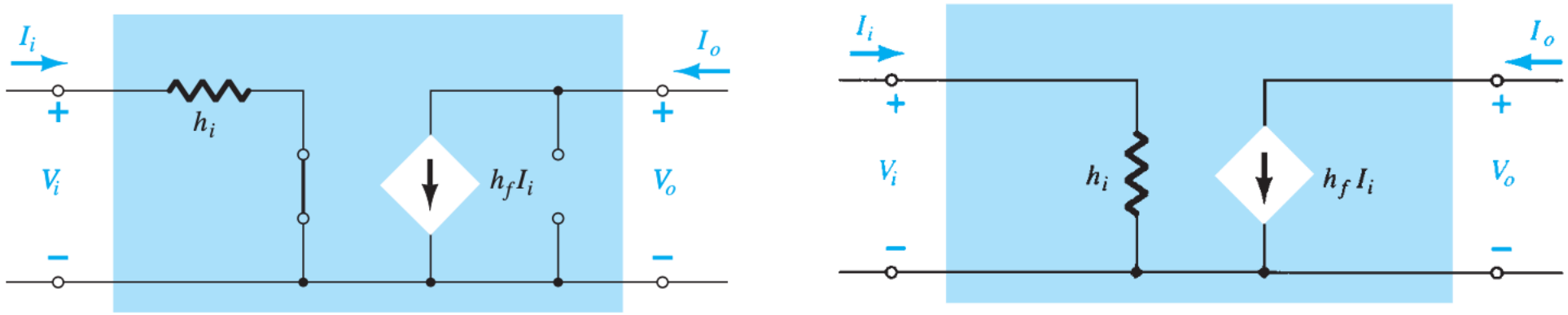
- Common-base configuration:





THE HYBRID EQUIVALENT MODEL (Cont…)

- Because h_r is normally a relatively small quality, its removal is approximated by $h_r = 0$ and $h_r V_o = 0$, resulting in a short-circuit equivalent for the feedback element.
- The resistance $1/h_o$ is often large enough to be ignored in comparison to a parallel load, permitting its replacement by an open-circuit equivalent for the CE and CB models



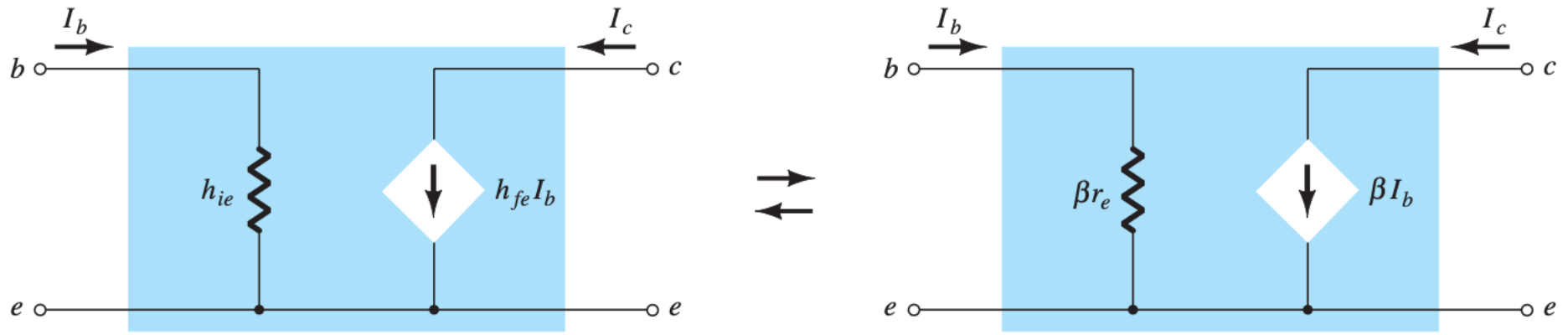
THE HYBRID EQUIVALENT MODEL (Cont…)

Hybrid versus r_e model

- CE

$$h_{ie} = \beta r_e$$

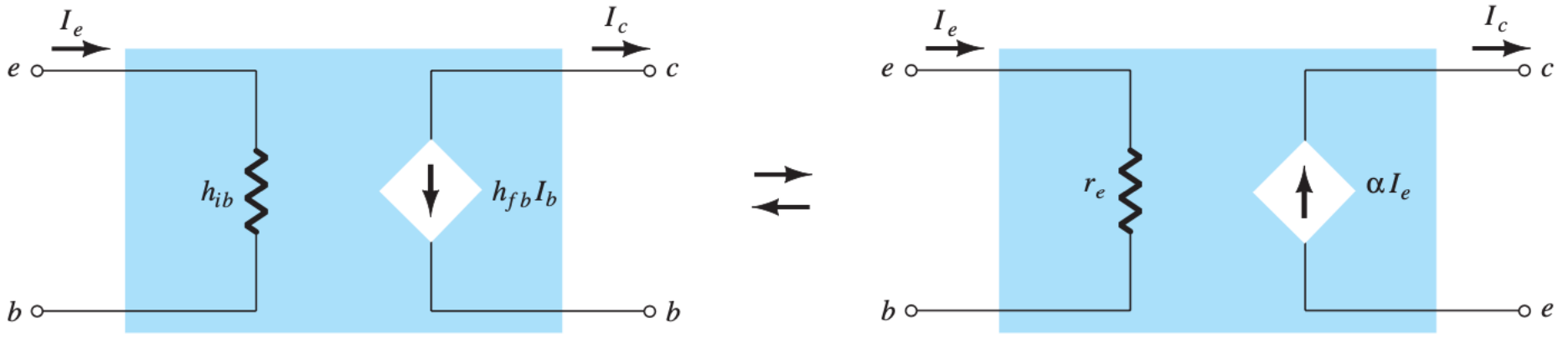
$$h_{fe} = \beta_{ac}$$



- CB

$$h_{ib} = r_e$$

$$h_{fb} = -\alpha \cong -1$$



THE HYBRID EQUIVALENT MODEL (Cont…)

- *Hybrid parameters for the 2N4400 transistor from its datasheet*

		Min.	Max.	
Input impedance (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz)	h_{ie}	0.5	7.5	kΩ
Voltage feedback ratio (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz)	h_{re}	0.1	8.0	$\times 10^{-4}$
Small-signal current gain (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz)	h_{fe}	20	250	—
Output admittance (I_C = 1 mA dc, V_{CE} = 10 V dc, f = 1 kHz)	h_{oe}	1.0	30	1 μS

Introduction of Feedback Amplifiers

The phenomenon of feeding a portion of the output signal back to the input circuit is known as feedback.

The effect results in a dependence between the output and the input and an effective control can be obtained in the working of the circuit. Feedback is of two types:

- **Positive Feedback**
- **Negative Feedback**

Introduction of Feedback Amplifiers (Cont…)

Positive or regenerate feedback:

- In positive feedback, the feedback energy (voltage or currents), is in phase with the input signal and thus aids it. Positive feedback increases gain of the amplifier also increases distortion, noise and instability.
- Because of these disadvantages, positive feedback is seldom employed in amplifiers. But the positive feedback is used in oscillators.

Introduction of Feedback Amplifiers (Cont...)

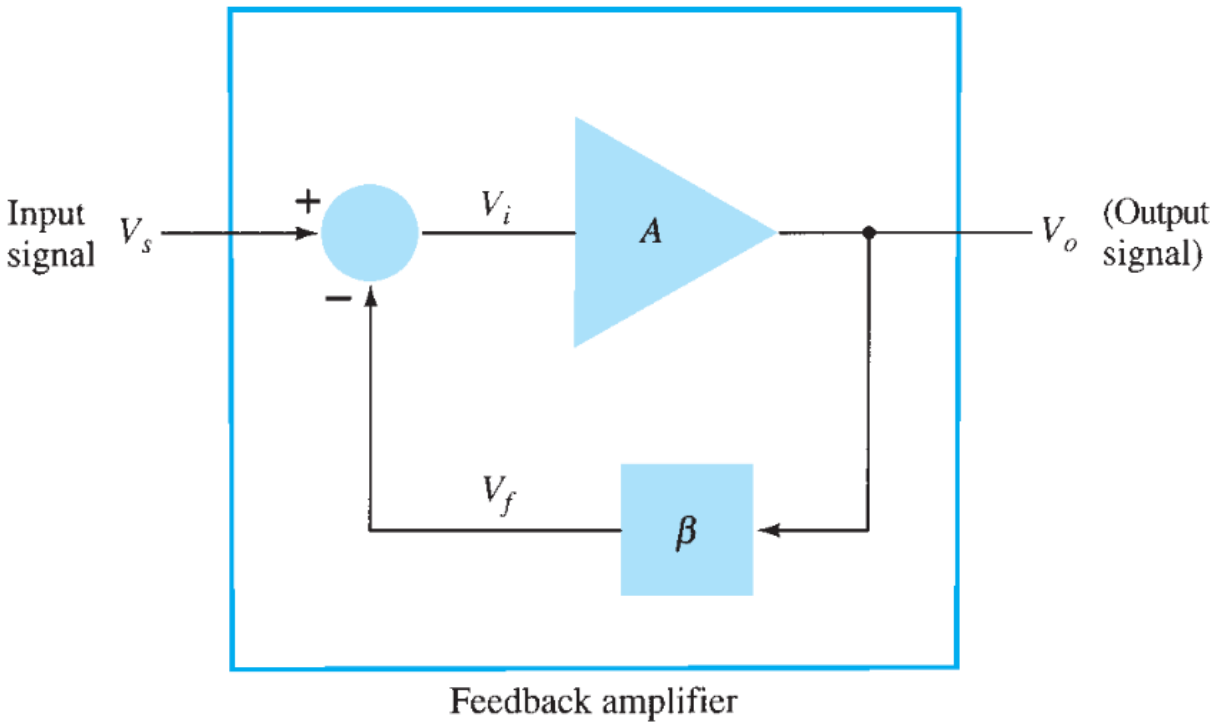
Negative or Degenerate feedback:

- In negative feedback, the feedback energy (voltage or current), is out of phase with the input signal and thus opposes it.
- Negative feedback reduces gain of the amplifier. It also reduce distortion, noise and instability.
- This feedback increases bandwidth and improves input and output impedances.
- Due to these advantages, the negative feedback is frequently used in amplifiers.

Comparison Between Positive and Negative Feed Back

No.	Negative Feedback	Positive Feedback
1	Feedback energy is out phase with their input signal	Feedback energy is in phase with the input signal
2	Gain of the amplifier decreases	Gain of the amplifier increases
3	Gain stability increases	Gain stability decreases
4	Noise and distortion decreases	Noise and distortion increases
5	Increase the band width	Decreases bandwidth
6	Used in amplifiers	Used in Oscillators

Principle of Feedback Amplifier



Simple block diagram of feedback amplifier

A feedback amplifier generally consists of two parts. They are the **amplifier (A)** and the **feedback circuit (β)**. The feedback circuit usually consists of resistors.

Principle of Feedback Amplifier (Cont…)

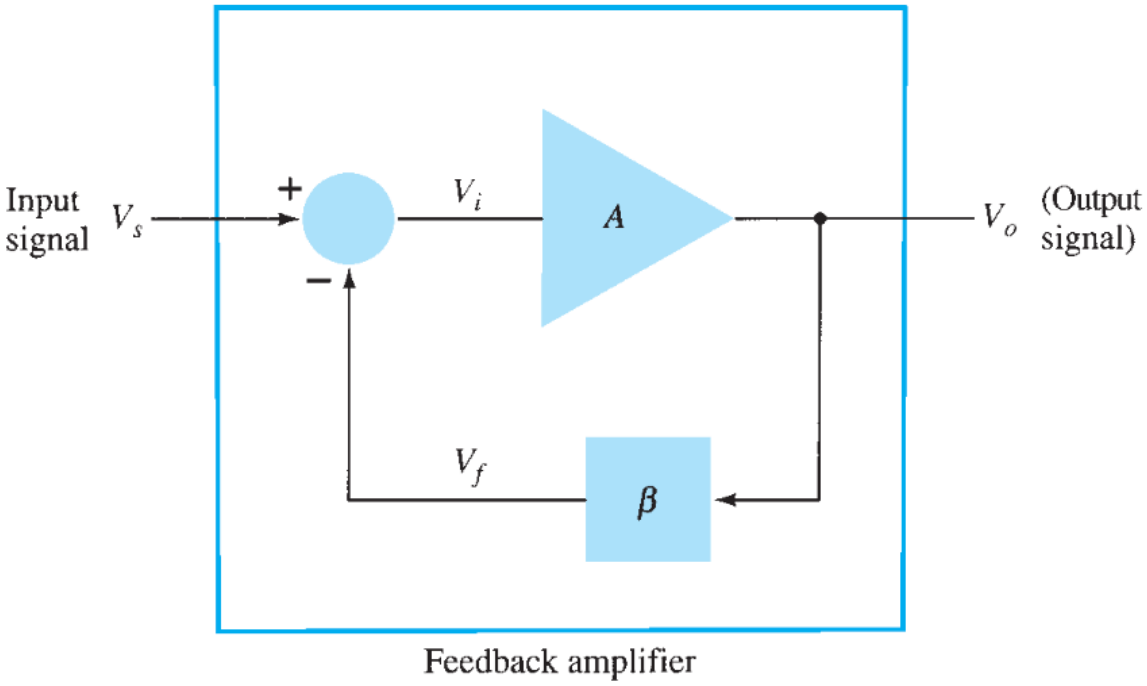
- The gain of the amplifier is represented as A. The gain of the amplifier is the ratio of output voltage V_o to the input voltage V_i . The feedback network extracts a voltage

$V_f = \beta V_o$ from the output Vo of the amplifier.

- This voltage is subtracted for negative feedback, from the signal voltage V_s. Now,

$$V_i = V_s - V_f = V_s - \beta V_o$$

- The quantity **$\beta = V_f / V_o$** is called as feedback ratio or feedback fraction.



Principle of Feedback Amplifier (Cont…)

- The output Vo must be equal to the input voltage (Vs - βVo) multiplied by the gain A of the amplifier. Hence,

$$(Vs-\beta Vo)A=Vo$$

$$AVs-A\beta Vo=Vo$$

$$AVs=Vo(1+A\beta)$$

$$Vo/Vs=A/(1+A\beta)$$

- Therefore, the gain of the amplifier with feedback is given by

$$A_f=A/(1+A\beta)$$

Feedback Connection Types

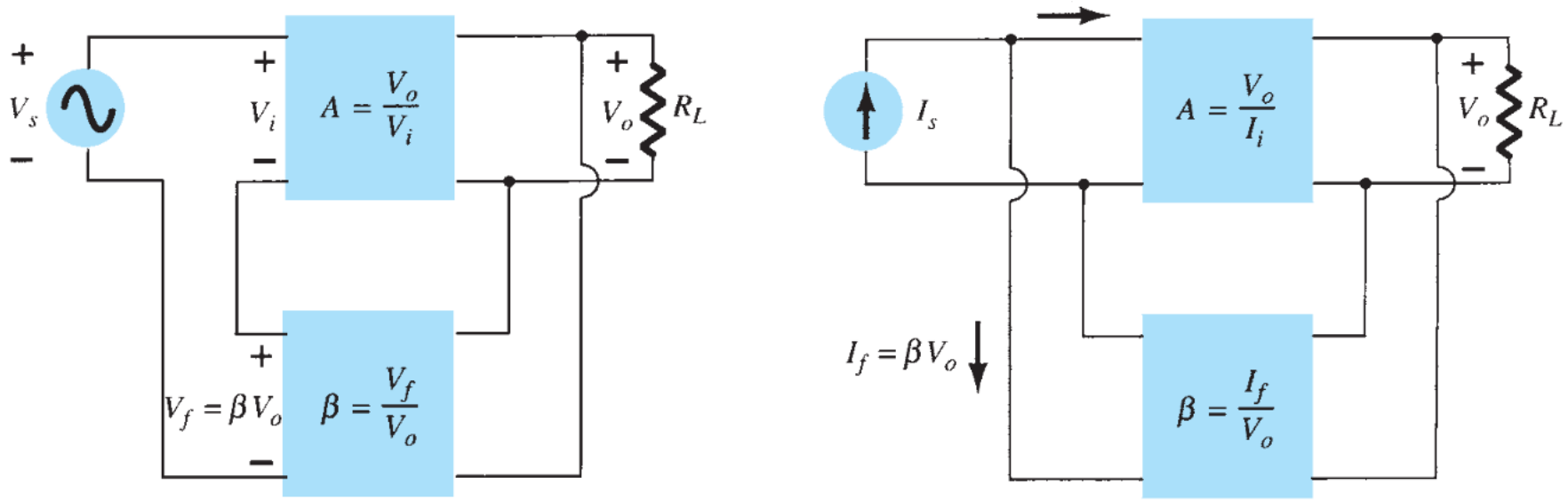
Both *voltage* and *current* can be fed back to the input either in *series* or *parallel.* Specifically, there can be:

1. Voltage-series feedback
2. Voltage-shunt feedback
3. Current-series feedback
4. Current-shunt feedback

- ***Voltage*** refers to connecting the output voltage as input to the feed-back network;
- ***Current*** refers to tapping off some output current through the feedback network;
- ***Series*** refers to connecting the feedback signal in series with the input signal voltage;
- ***Shunt*** refers to connecting the feedback signal in shunt (parallel) with an input current source.

Feedback Connection Types (Cont…)

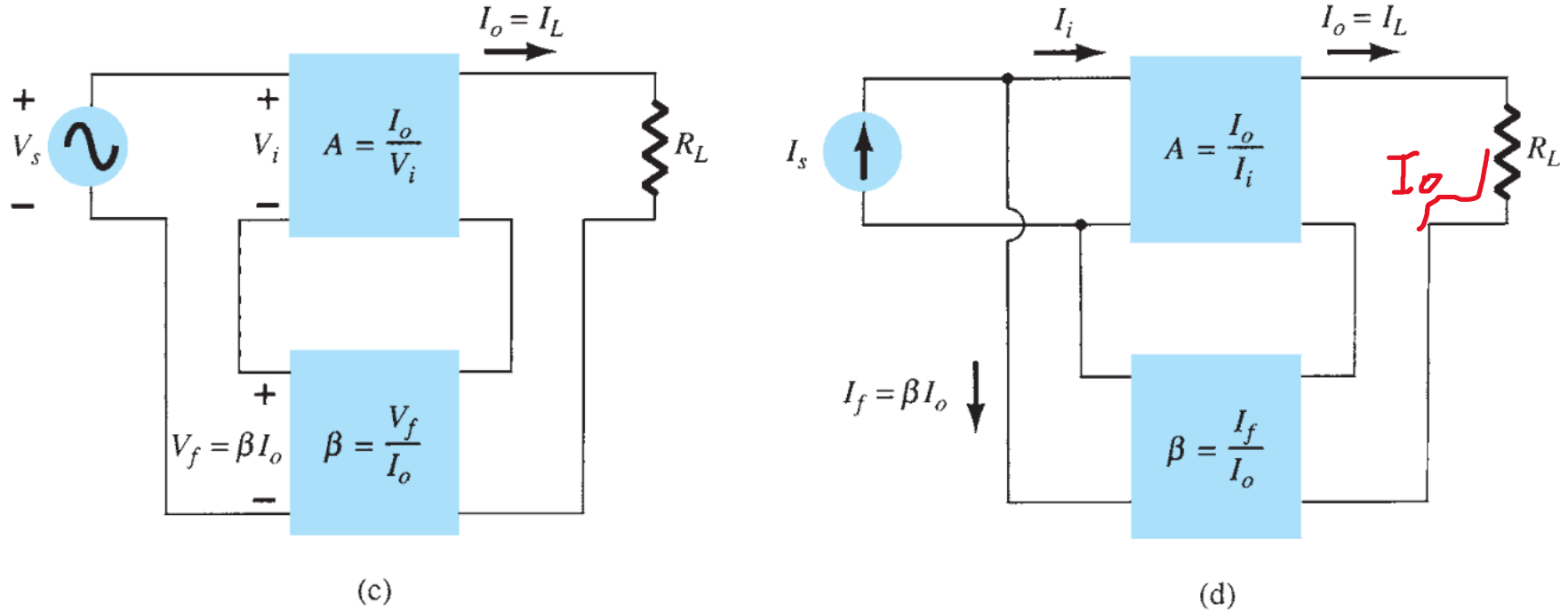
- Series feedback connections tend to *increase* the input resistance
- Shunt feed-back connections tend to *decrease* the input resistance
- Voltage feedback tends to *decrease* the output impedance



(a) voltage-series feedback, $A_f = V_o/V_s$; (b) voltage-shunt feedback, $A_f = V_o/I_s$;

Feedback Connection Types (Cont…)

- Series feedback connections tend to *increase* the input resistance
- Shunt feed-back connections tend to *decrease* the input resistance
- Current feedback tends to *increase* the output impedance



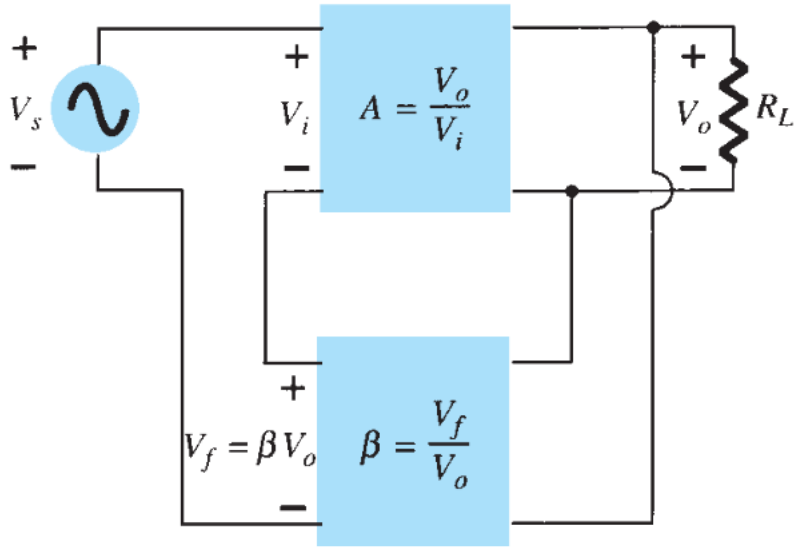
(c) current-series feedback, $A_f = I_o/V_s$; (d) current-shunt feedback, $A_f = I_o/I_s$.

Gain with Feedback

The gain without feedback, *A*, is that of the amplifier stage. With feedback **β**, the overall gain of the circuit is reduced by a factor (1 + **β***A*)

		Voltage-Series	Voltage-Shunt	Current-Series	Current-Shunt
Gain without feedback	A	$\frac{V_o}{V_i}$	$\frac{V_o}{I_i}$	$\frac{I_o}{V_i}$	$\frac{I_o}{I_i}$
Feedback	β	$\frac{V_f}{V_o}$	$\frac{I_f}{V_o}$	$\frac{V_f}{I_o}$	$\frac{I_f}{I_o}$
Gain with feedback	A_f	$\frac{V_o}{V_s}$	$\frac{V_o}{I_s}$	$\frac{I_o}{V_s}$	$\frac{I_o}{I_s}$

Gain as in Voltage-Series Feedback



$$A = \frac{V_o}{V_s} = \frac{V_o}{V_i}$$

If a feedback signal V_f is connected in series with the input, then

$$V_i = V_s - V_f$$

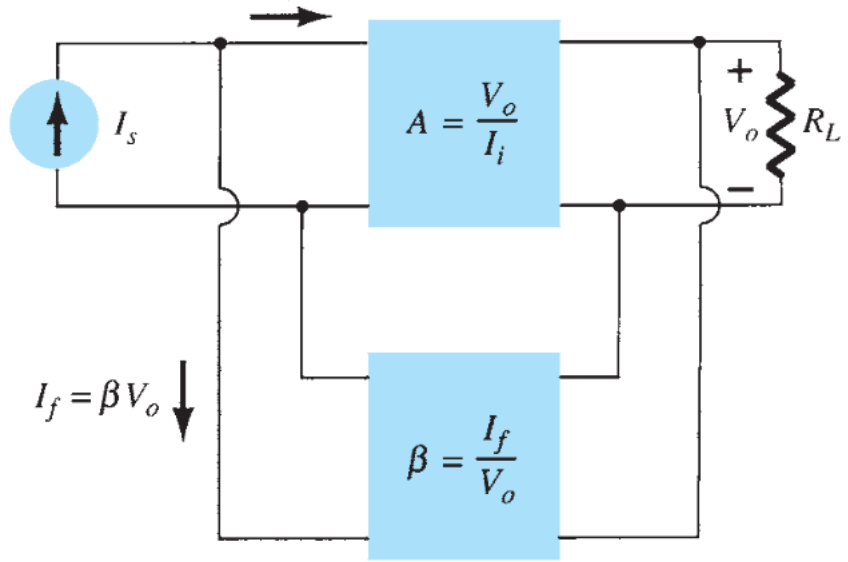
Since $$V_o = AV_i = A(V_s - V_f) = AV_s - AV_f = AV_s - A(\beta V_o)$$

then $$(1 + \beta A)V_o = AV_s$$

so that the overall voltage gain *with* feedback is

$$A_f = \frac{V_o}{V_s} = \frac{A}{1 + \beta A}$$

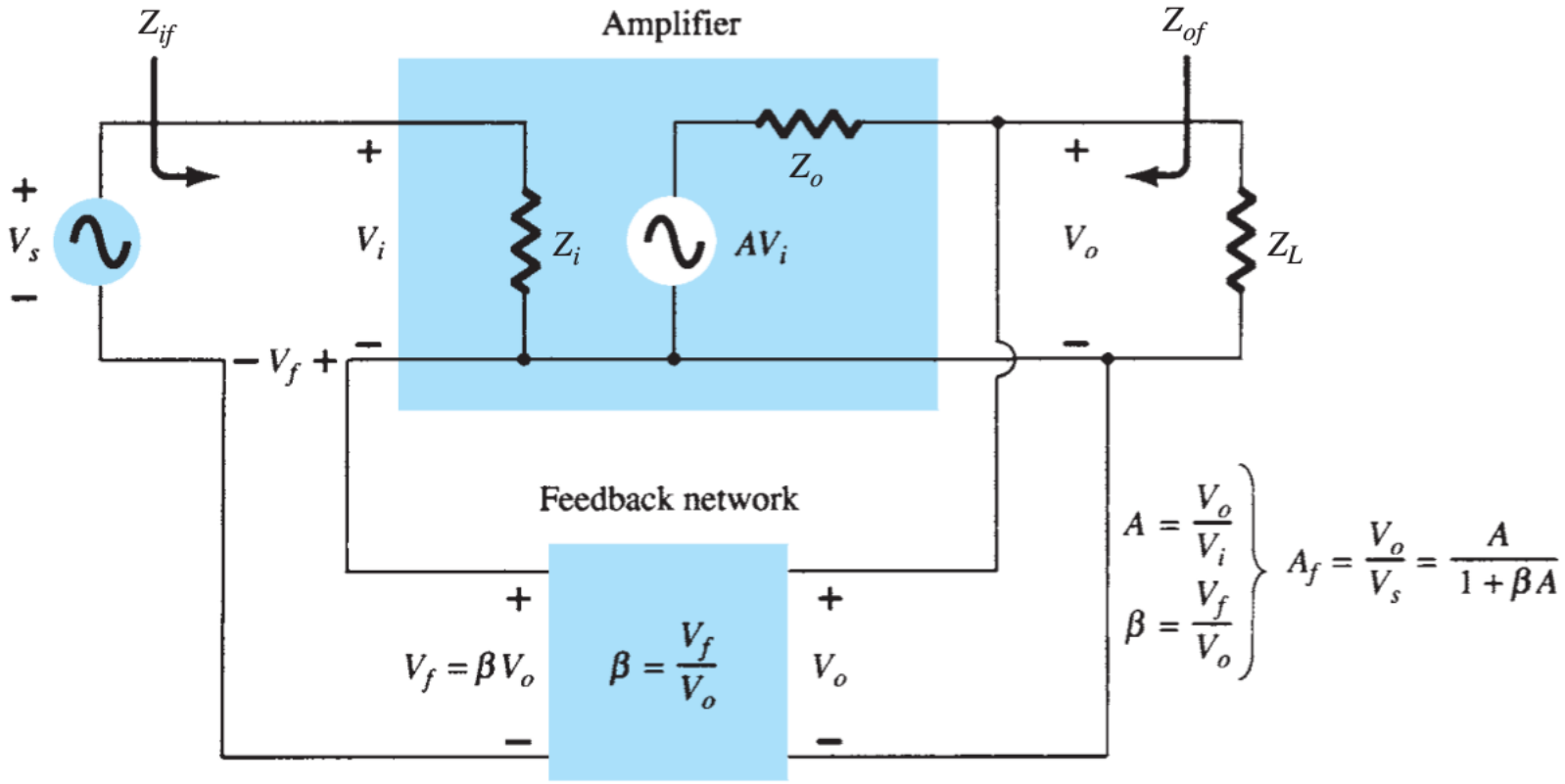
Gain as in Voltage-Shunt Feedback



$$A_f = \frac{V_o}{I_s} = \frac{A\,I_i}{I_i + I_f} = \frac{A\,I_i}{I_i + \beta V_o} = \frac{A\,I_i}{I_i + \beta A\,I_i}$$

$$\boxed{A_f = \frac{A}{1 + \beta A}}$$

Input Impedance of Voltage-Series Feedback



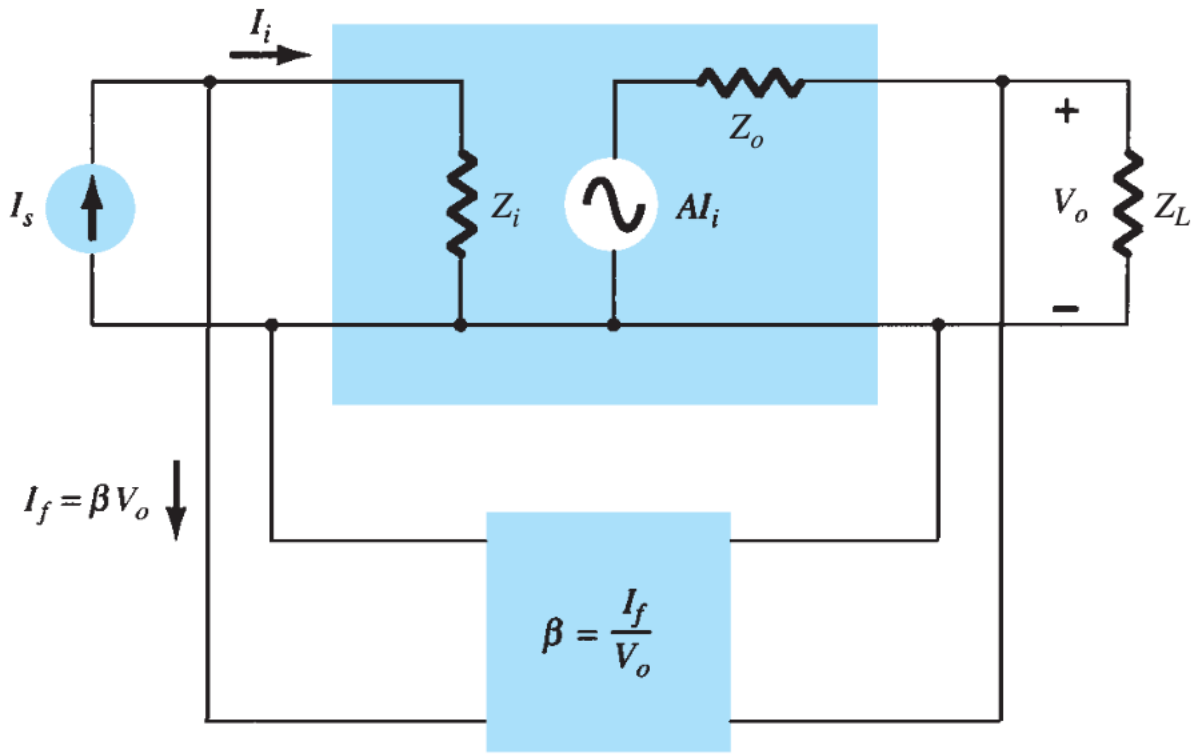
$$I_i = \frac{V_i}{Z_i} = \frac{V_s - V_f}{Z_i} = \frac{V_s - \beta V_o}{Z_i} = \frac{V_s - \beta A V_i}{Z_i}$$

$$Z_i = V_s - \beta A V_i$$

$$V_s = I_i Z_i + \beta A V_i = I_i Z_i + \beta A I_i Z_i$$

$$\boxed{Z_{if} = \frac{V_s}{I_i} = Z_i + (\beta A) Z_i = Z_i(1 + \beta A)}$$

Input Impedance of Voltage-Shunt Feedback



$$Z_{if} = \frac{V_i}{I_s} = \frac{V_i}{I_i + I_f} = \frac{V_i}{I_i + \beta V_o}$$

$$= \frac{V_i/I_i}{I_i/I_i + \beta V_o/I_i}$$

$$\boxed{Z_{if} = \frac{Z_i}{1 + \beta A}}$$

Output Impedance of Voltage-Series Feedback

$$V = IZ_o + AV_i$$

For $V_s = 0$, $V_i = -V_f$

so that $V = IZ_o - AV_f = IZ_o - A(\beta V)$

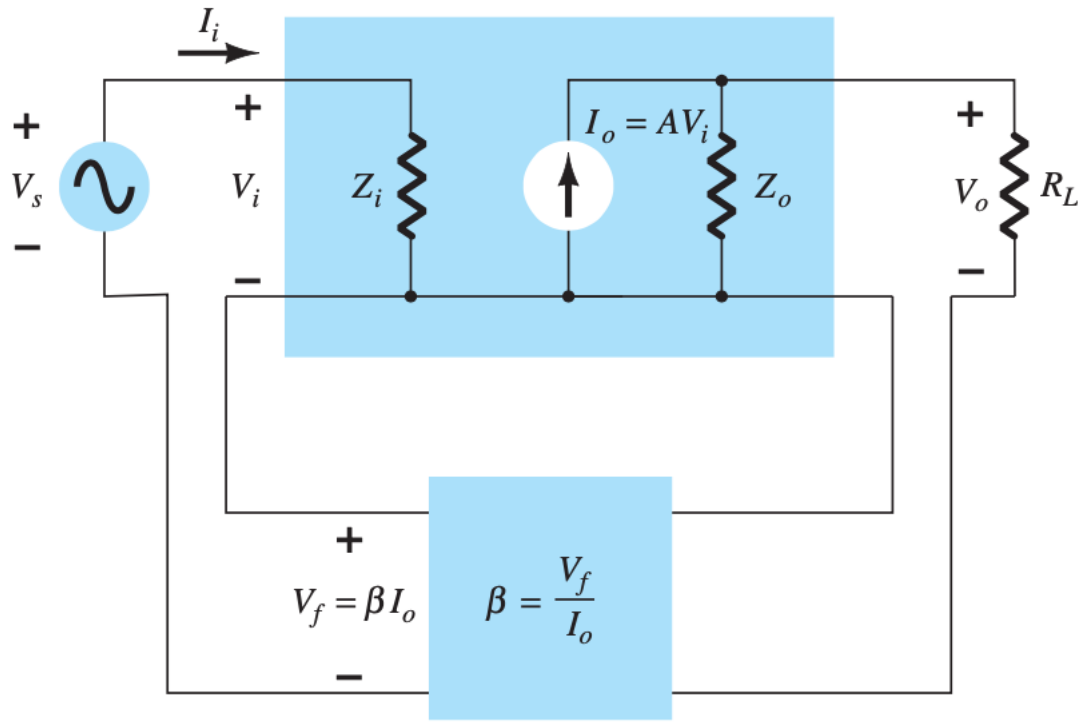
Rewriting the equation as

$$V + \beta AV = IZ_o$$

allows solving for the output impedance with feedback:

$$\boxed{Z_{of} = \frac{V}{I} = \frac{Z_o}{1 + \beta A}}$$

Output Impedance of Current-Series Feedback



$$V_i = V_f$$

$$I = \frac{V}{Z_o} - AV_i = \frac{V}{Z_o} - AV_f = \frac{V}{Z_o} - A\beta I$$

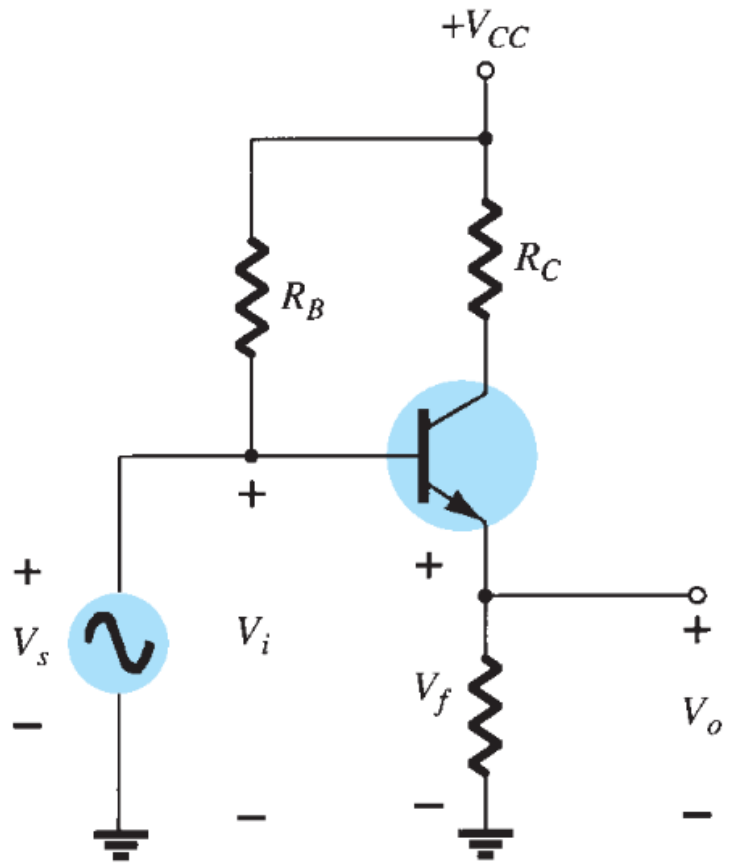
$$Z_o(1 + \beta A)I = V$$

$$\boxed{Z_{of} = \frac{V}{I} = Z_o(1 + \beta A)}$$

Effect of Feedback Connection on Input and Output Impedance

Voltage-Series	Current-Series	Voltage-Shunt	Current-Shunt
Z_{if} $Z_i(1 + \beta A)$ (increased)	$Z_i(1 + \beta A)$ (increased)	$\frac{Z_i}{1 + \beta A}$ (decreased)	$\frac{Z_i}{1 + \beta A}$ (decreased)
Z_{of} $\frac{Z_o}{1 + \beta A}$ (decreased)	$Z_o(1 + \beta A)$ (increased)	$\frac{Z_o}{1 + \beta A}$ (decreased)	$Z_o(1 + \beta A)$ (increased)

Voltage-series feedback circuit (emitter-follower)



$$A = \frac{V_o}{V_s} = \frac{h_{fe} I_b R_E}{V_s} = \frac{h_{fe} R_E (V_s/h_{ie})}{V_s} = \frac{h_{fe} R_E}{h_{ie}}$$

$$\beta = \frac{V_f}{V_o} = 1$$

The operation with feedback then provides that

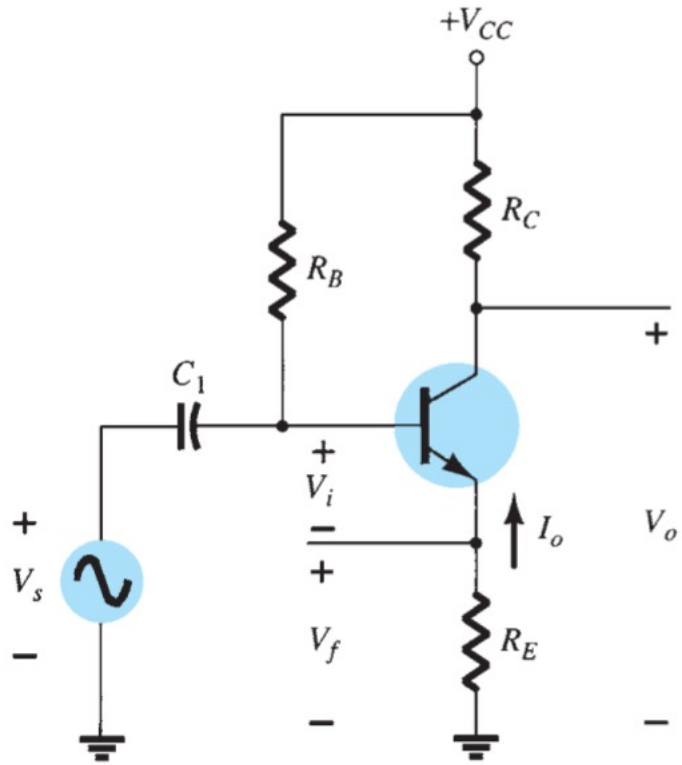
$$A_f = \frac{V_o}{V_s} = \frac{A}{1 + \beta A} = \frac{h_{fe} R_E / h_{ie}}{1 + (1)(h_{fe} R_E / h_{ie})}$$

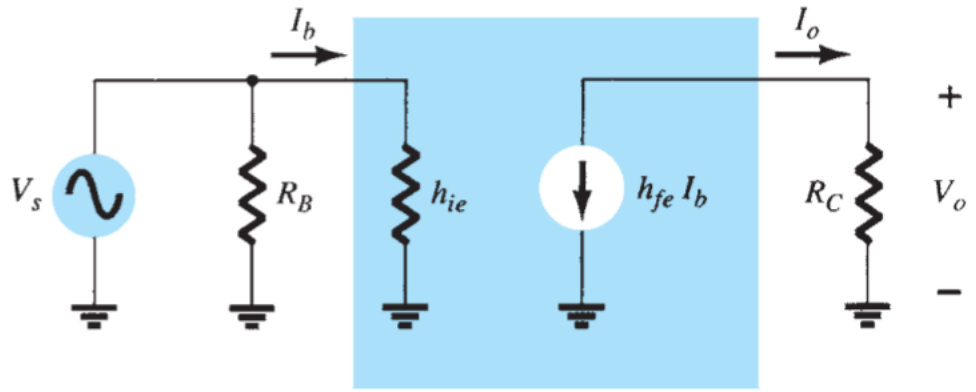
$$= \frac{h_{fe} R_E}{h_{ie} + h_{fe} R_E}$$

For $h_{fe} R_E \gg h_{ie}$,

$$A_f \cong 1$$

Current-series feedback





$$A = \frac{I_o}{V_i} = \frac{-I_b h_{fe}}{I_b h_{ie} + R_E} = \frac{-h_{fe}}{h_{ie} + R_E}$$

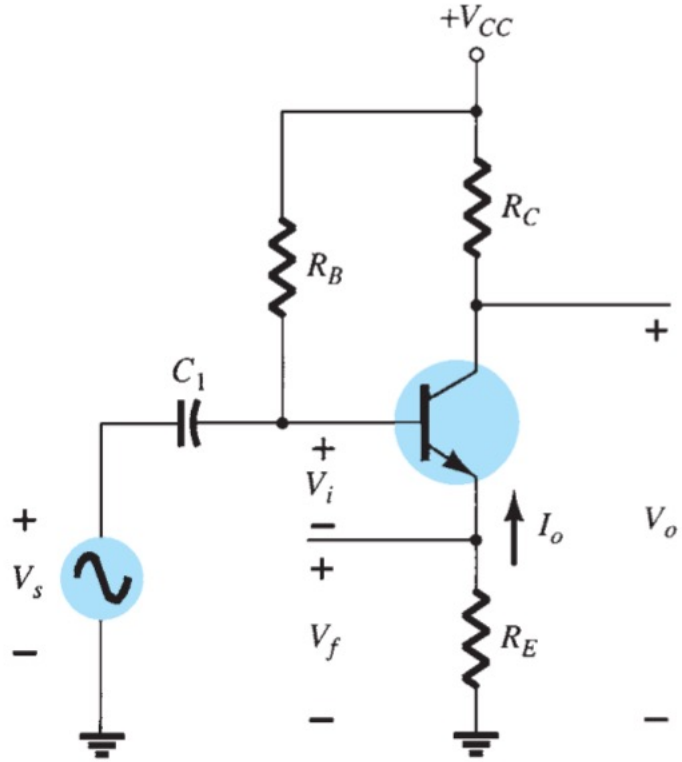
$$\beta = \frac{V_f}{I_o} = \frac{-I_o R_E}{I_o} = -R_E$$

The input and output impedances are, respectively,

$$Z_i = R_B \| (h_{ie} + R_E) \cong h_{ie} + R_E$$

$$Z_o = R_C$$

Current-series feedback



With Feedback

$$A_f = \frac{I_o}{V_s} = \frac{A}{1 + \beta A} = \frac{-h_{fe}/h_{ie}}{1 + (-R_E)\left(\dfrac{-h_{fe}}{h_{ie} + R_E}\right)} \cong \frac{-h_{fe}}{h_{ie} + h_{fe}R_E}$$

The input and output impedances are calculated as specified in Table 14.2:

$$Z_{if} = Z_i(1 + \beta A) \cong h_{ie}\left(1 + \frac{h_{fe}R_E}{h_{ie}}\right) = h_{ie} + h_{fe}R_E$$

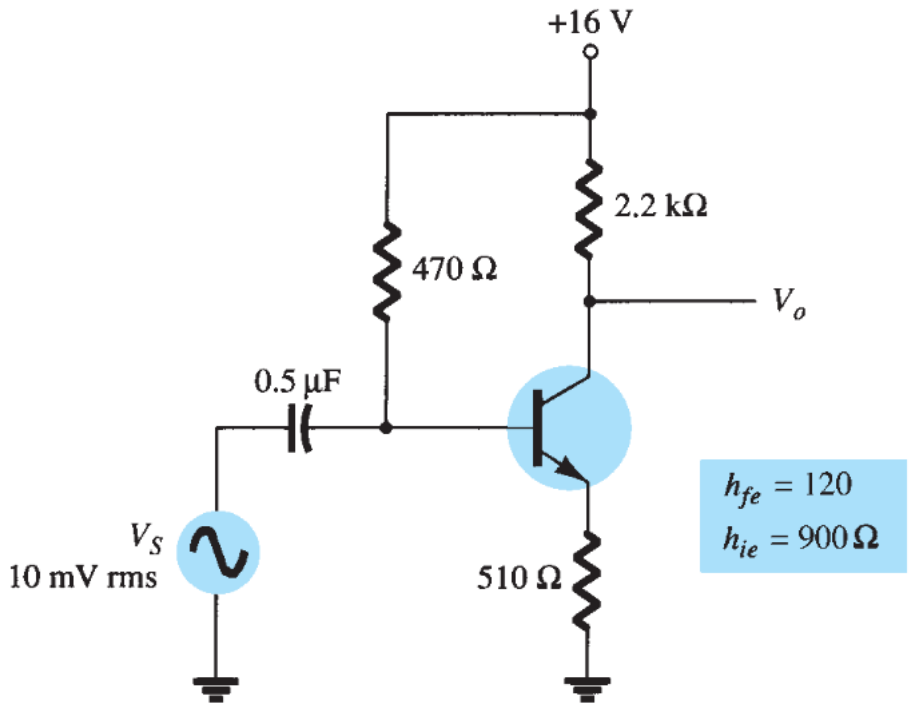
$$Z_{of} = Z_o(1 + \beta A) = R_C\left(1 + \frac{h_{fe}R_E}{h_{ie}}\right)$$

The voltage gain A with feedback is

$$A_{vf} = \frac{V_o}{V_s} = \frac{I_o R_C}{V_s} = \left(\frac{I_o}{V_s}\right)R_C = A_f R_C \cong \frac{-h_{fe}R_C}{h_{ie} + h_{fe}R_E}$$

Exercise 1

- Calculate the voltage gain of the circuit:



Solution: Without feedback,

$$A = \frac{I_o}{V_i} = \frac{-h_{fe}}{h_{ie} + R_E} = \frac{-120}{900 + 510} = -0.085$$

$$\beta = \frac{V_f}{I_o} = -R_E = -510$$

The factor $(1 + \beta A)$ is then

$$1 + \beta A = 1 + (-0.085)(-510) = 44.35$$

The gain with feedback is then

$$A_f = \frac{I_o}{V_s} = \frac{A}{1 + \beta A} = \frac{-0.085}{44.35} = -1.92 \times 10^{-3}$$

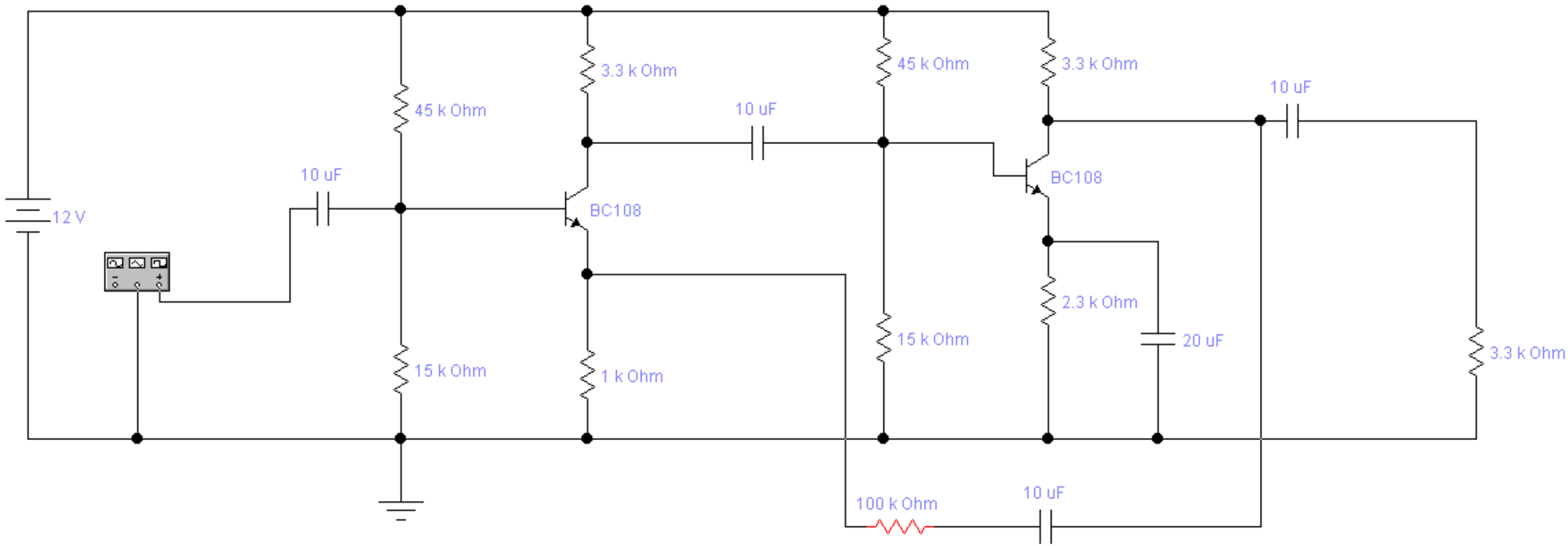
and the voltage gain with feedback A_{vf} is

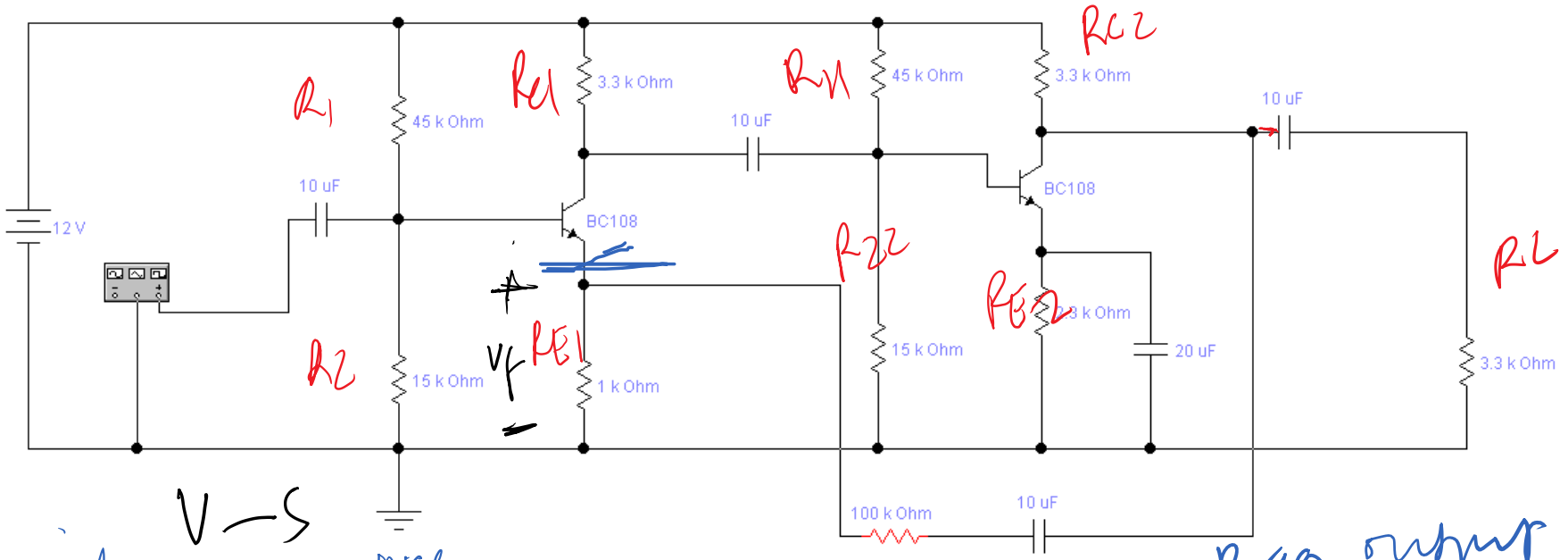
$$A_{vf} = \frac{V_o}{V_s} = A_f R_C = (-1.92 \times 10^{-3})(2.2 \times 10^3) = \mathbf{-4.2}$$

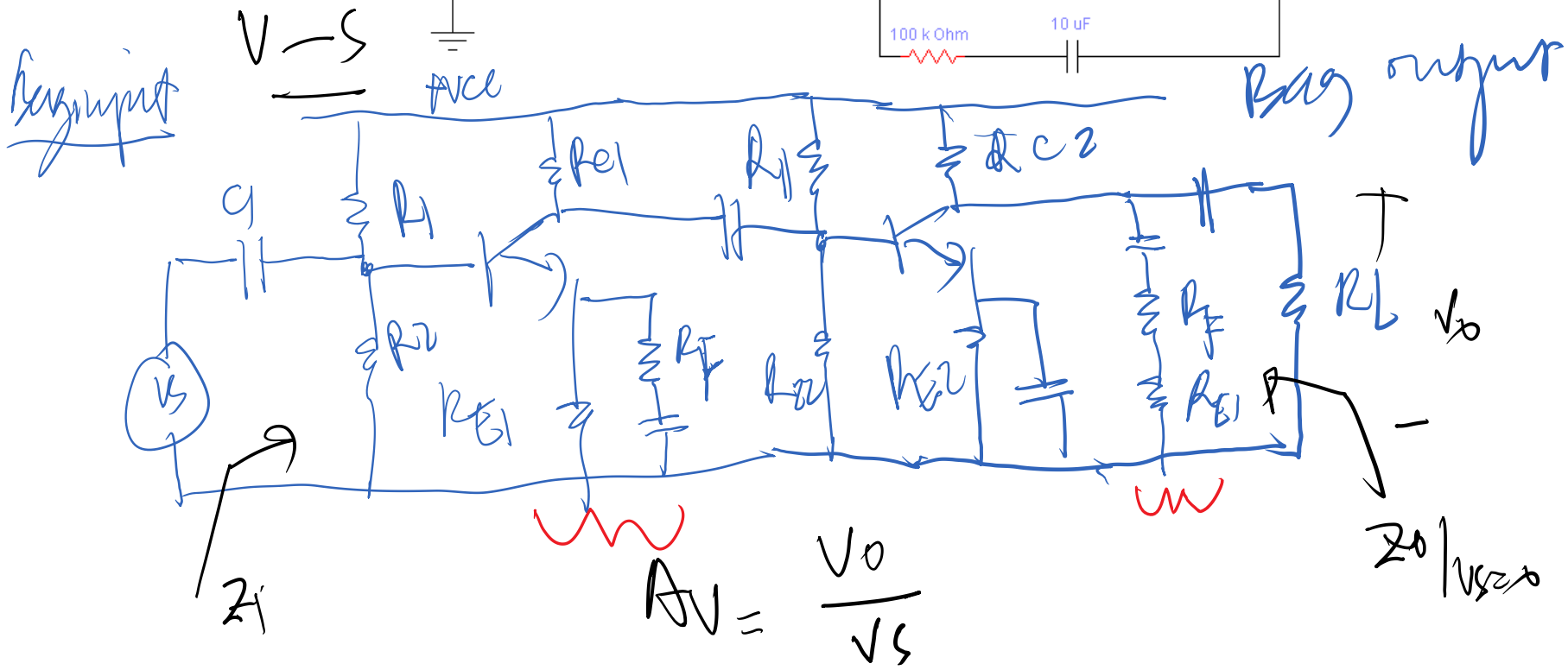
Without feedback ($R_E = 0$), the voltage gain is

$$A_v = \frac{-R_C}{r_e} = \frac{-2.2 \times 10^3}{7.5} = \mathbf{-293.3}$$

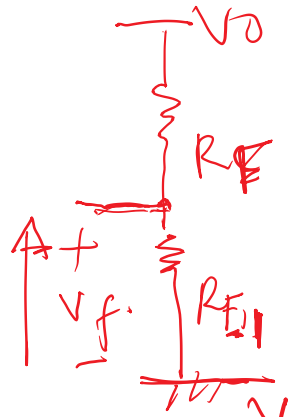
Exercise 2


12 V
10 uF
45 k Ohm
15 k Ohm
3.3 k Ohm
10 uF
BC108
1 k Ohm
45 k Ohm
15 k Ohm
3.3 k Ohm
BC108
2.3 k Ohm
20 uF
10 uF
3.3 k Ohm
100 k Ohm
10 uF

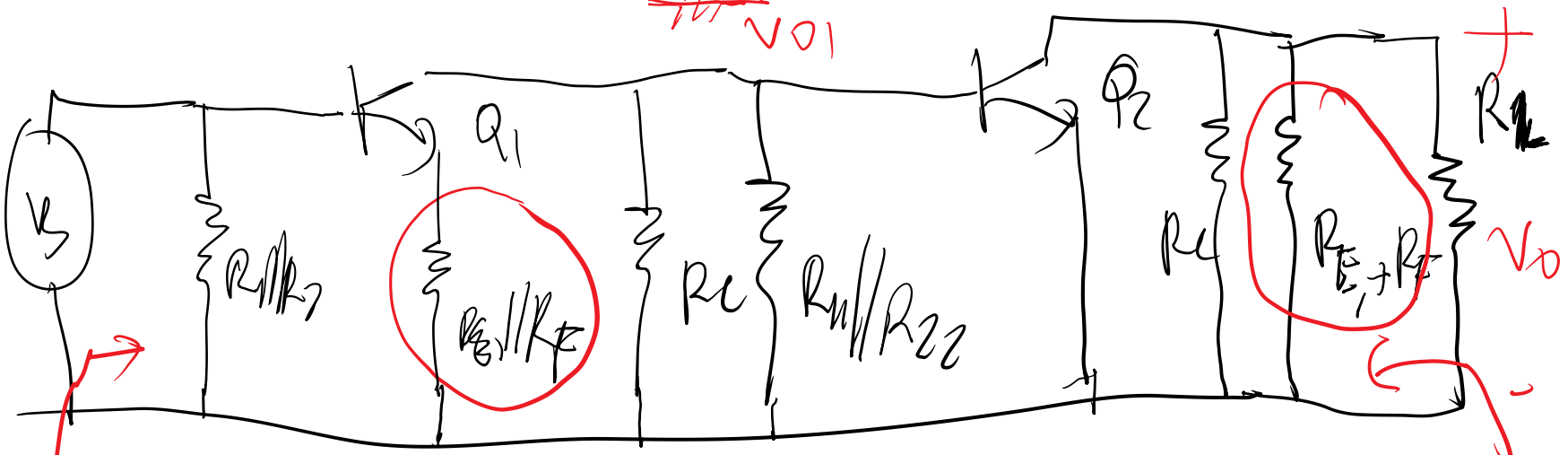

12 V
10 uF
45 k Ohm
15 k Ohm
3.3 k Ohm
1 k Ohm
BC108
10 uF
45 k Ohm
15 k Ohm
3.3 k Ohm
BC108
20 uF
10 uF
3.3 k Ohm
100 k Ohm
10 uF


Vcc
C1
R1
R2
RE1
RF
RC1
RC2
RL
Vo
Vs
Zi
Zo

$$\beta = \frac{V_f}{V_0} = \frac{R_{E1}}{R_{E1} + R_F}$$



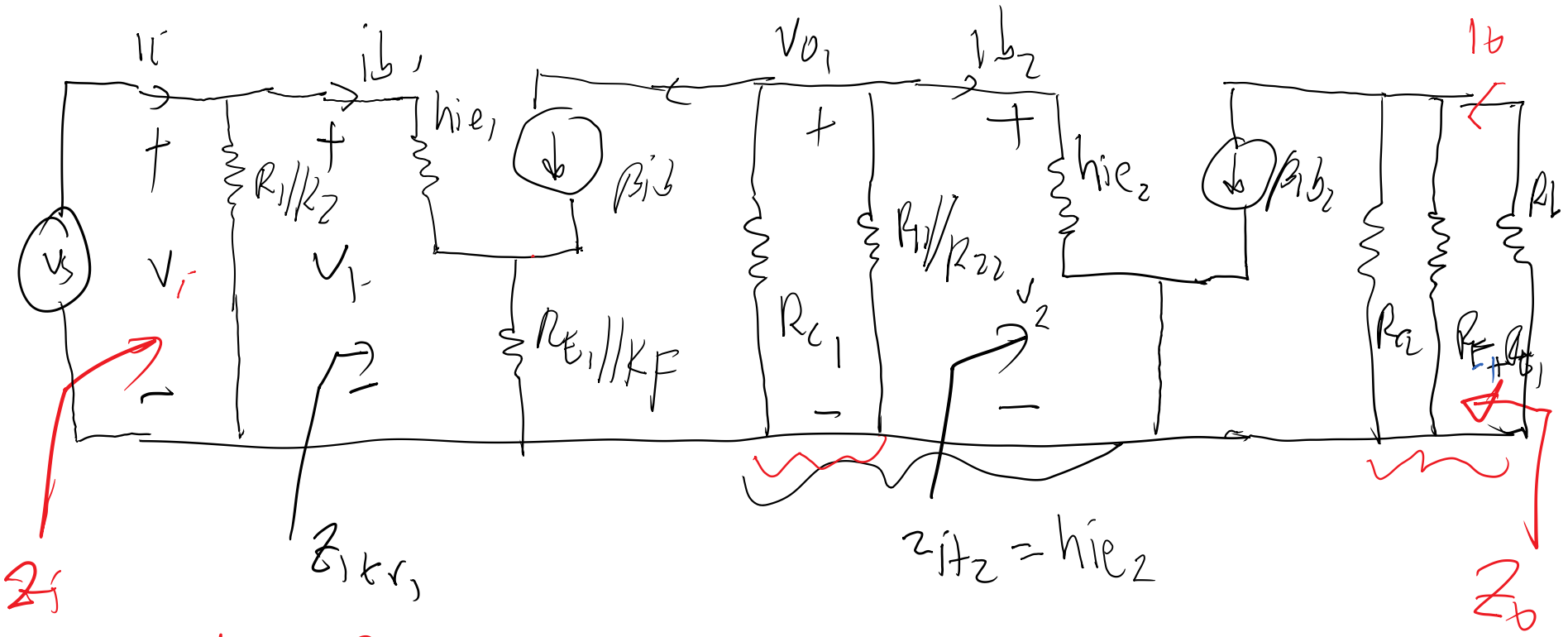
$$\beta = \frac{1}{1+100} = 0.011$$



Z_i

$$A_V = \frac{V_0}{V_s} = \frac{V_0}{V_{01}} \cdot \frac{V_{01}}{V_s}$$

$Z_0 |_{V_s}$



$$Z_i = (R_1 \| R_2) \| Z_{if_1}$$

$$Z_{if_1} = \frac{V_1}{i_b} = \frac{h_{ie_1} i_b + (\beta+1) R_{E_1} \| R_F}{i_b}$$

$$= h_{ie_1} + (\beta+1) R_{E_1} \| R_F$$

$$Z_i = (R_1 \| R_2) \| (h_{ie_1} + (\beta+1) R_{E_1} \| R_F)$$

$$Z_{if} = Z_i (1 + A\beta)$$

$$Z_o = R_{C_2} \| (R_{B_2} + R_F)$$

$$Z_o' = Z_o \| R_L$$

$$A_V = \frac{V_o}{V_s} = \frac{V_o}{V_{o1}} \cdot \frac{V_{o1}}{V_s}$$

$$\frac{V_o}{V_{o1}} = -\frac{R_{C2} \| (R_{E1} + R_f) \| R_L \cdot \beta_2 \cdot I_{b2}}{I_{b2} \cdot h_{ie2}}$$

$$= -\frac{R_{C2} \| (R_{E1} + R_f) \| R_L \cdot \beta_2 \cdot I_{b2}}{I_{b2} \cdot h_{ie2} \, r_{e2}}$$

$$= -\frac{R_{C2} \| (R_{E1} + R_f) \| R_L}{r_{e2}}$$

$$r_{e2} = \frac{26\,mV}{I_{E2}\,mA}$$

$$\frac{V_{01}}{V_s} = \frac{-R_{C1} \| (R_{11} \| R_{22}) \| hie_2 \cdot \beta_2 \cdot ib_2}{ib_1 \cdot hie_1 + ie_1 (R_{E1} + R_f)}$$

hie_2

$$= \frac{-R_{C1} \| (R_{11} \| R_{22}) \| hie_2 \cdot \beta_2 \cdot ib_1}{ib_1 (\beta+1) re + ib_1 (\beta+1)(R_{E1} + R_f)}$$

$$\frac{V_0}{V_s} = \frac{-R_{C1} \| (R_{11} \| R_{22}) \| hie_2}{re_1 + (R_{E1} \| R_f)} \cdot \cdots\cdots\cdots$$

$$re = \frac{26mV}{I_E} \; ; \; hie = \beta re$$

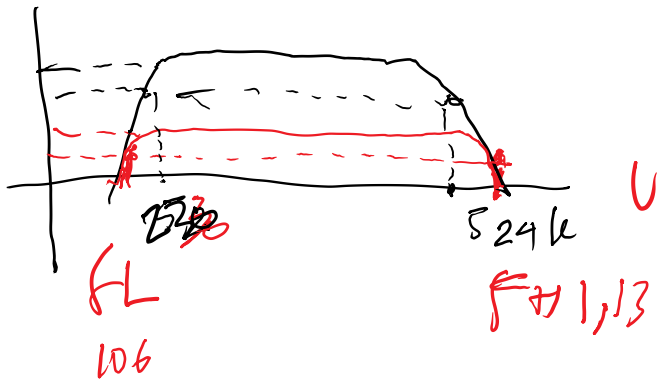
Hasil simulasi Rangkaian dengan feedback.

A_f : 50 X

Z_{if} : 12 kΩ

Z_{of} : 1080 Ω

BW_f : 1,13 MHz – 106 Hz



Rangkaian ekivalen tipe fb

A = 113 X

Z_i = 10,1 k

Z_o = 2,713 kΩ

BW = 524kHz – 230hz

$A_f = \frac{A}{1+A\beta}$; $\beta = \frac{1}{1+100}$ ≅ 0,01

$= \frac{113}{1+113.0,01} = \frac{113}{1+1,13} = 53$

$Z_f = Z_i(2,13) = 10,1 \times 2,13 = 29\,Z$

$Z_{of} = 2713\,\Omega / 2,13 = 1,273\,\Omega$